

DevOps Documentation

Program 6: Deploy an Automated Build Pipeline using Docker Hub

Program-6.2

—> Dockerfile

—> app.py

—> requirements.txt

S1. Create a simple Dockerfile

nano Dockerfile

Create a simple app.py file

nano app.py

Create a simple requirements.txt file

nano requirements.txt

Build an image

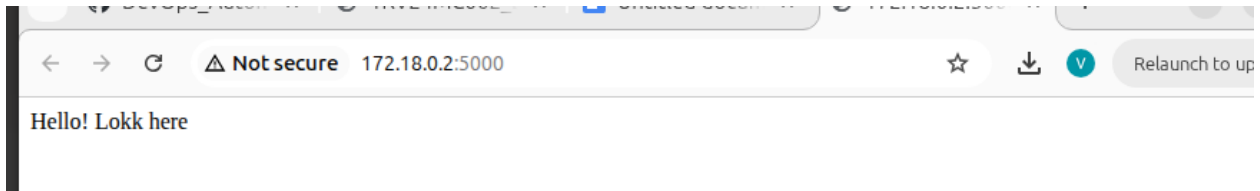
docker build -t program6.2 .

```
● 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program-6.1$ docker build -t program6.1 .
[+] Building 2.4s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 192B
=> [internal] load metadata for docker.io/library/python:latest
=> [auth] library/python:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:latest@sha256:6d58c1a9444bc2664f0fa20c43a592fcd2698eb9a9c32257516538a2746c19a
=> [internal] load build context
=> => transferring context: 716B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:d7ca492591668fbedb0a94dfc71beba4fe2ed04b15a4d5a4561fb72bd120684
=> => naming to docker.io/library/program6.1
○ 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program-6.1$
```

Run the container

docker run -p 5000:5000 program6.2

```
○ 1rv24mc113_varun@varun-srivastava:~/DevOpslab/program-6.1$ docker run -p 5000:5000 program6.1
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.18.0.2:5000
Press CTRL+C to quit
█
```



S2. Tag the created image for Docker Hub

```
docker tag program6.2 varunsrivastava15/program6.2:latest
```

S3. Push the Image to docker hub

```
docker login -u varunsrivastava15
```

```
docker push varunsrivastava15/program6.2:latest
```

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program-6.1$ docker tag program6.1 varunsrivastava15/program6.1:latest
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program-6.1$ docker login -u varunsrivastava15

Info → A Personal Access Token (PAT) can be used instead.
To create a PAT, visit https://app.docker.com/settings
```

Password:

Login Succeeded

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program-6.1$ docker push varunsrivastava15/program6.1:latest
The push refers to repository [docker.io/varunsrivastava15/program6.1]
d4fdb38fdeee: Pushed
b2b6150ea99e: Pushed
47b971bb534f: Pushed
430339066c29: Pushed
aff6c5c1ab5c: Mounted from library/python
c2bb27004f5b: Mounted from library/python
76f0946f5ec2: Mounted from library/python
3e98cfd782c: Mounted from library/python
3a201f29126f: Mounted from library/python
68e78ea459e3: Mounted from library/python
14aa62b73a6f: Mounted from library/python
latest: digest: sha256:dcdb9f25631967285504e60234714f366999f2f9c53dd5662821a3cfac6767c3 size: 2627
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program-6.1$
```

S4. Tag the Image for GitHub Container Registry (GHCR)

```
docker tag program6.2 ghcr.io/varunvssrivastava/program6.2:latest
```

Push the Image to GHCR

```
export GITHUB_PAT="....."
```

```
echo $GITHUB_PAT | docker login ghcr.io -u varunvssrivastava --password-stdin
```

Push the image

```
docker push ghcr.io/varunvssrivastava/program6.2:latest
```

```
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program-6.1$ echo $GITHUB_PAT | docker login ghcr.io -u varunvssrivastava --password-stdin
WARNING! Your credentials are stored unencrypted in '/home/varun-srivastava/.docker/config.json'.
Configure a credential helper to remove this warning. See
https://docs.docker.com/go/credential-store/

Login Succeeded
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program-6.1$ docker push ghcr.io/varunvssrivastava/program6.1:latest
The push refers to repository [ghcr.io/varunvssrivastava/program6.1]
d4fdb38fdeee: Pushed
b2b6150ea99e: Pushed
47b971bb534f: Pushed
430339066c29: Pushed
aff6c5c1ab5c: Pushed
c2bb27004f5b: Pushed
76f0946f5ec2: Pushed
3e98cfd782c: Pushed
3a201f29126f: Pushed
68e78ea459e3: Pushed
14aa62b73a6f: Pushed
latest: digest: sha256:c3e557e8cc0e281cc2f955d394c543a6a6d429dcdb1411062a57b49ea5c04c2c size: 2627
1rv24mc113_varun@varun-srivastava:~/DevOpslab/program-6.1$
```

S5. Push source code to github

```
git init
```

```
git branch -M main
```

```
git remote add origin https://github.com/varunvssrivastava/program6.2.git
```

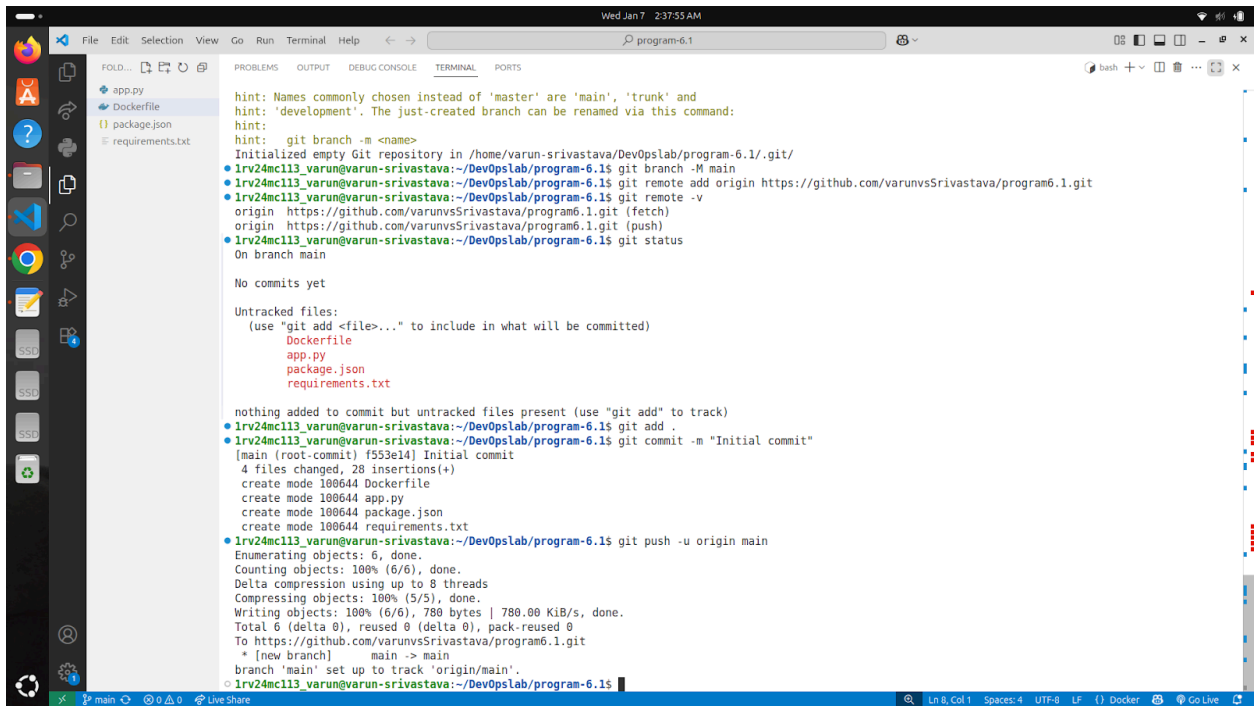
```
git remote -v
```

```
git status
```

```
git add .
```

```
git commit -m "Initial Commit"
```

```
git push -u origin main
```



```
Wed Jan 7 2:37:55 AM
program-6.1

FOLD...  app.py
          Dockerfile
          package.json
          requirements.txt

PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/varun-srivastava/DevOpsLab/program-6.1/.git/
• lrv24mc113_varun@varun-srivastava:~/DevOpsLab/program-6.1$ git branch -M main
• lrv24mc113_varun@varun-srivastava:~/DevOpsLab/program-6.1$ git remote add origin https://github.com/varunvssrivastava/program6.1.git
• lrv24mc113_varun@varun-srivastava:~/DevOpsLab/program-6.1$ git remote -v
origin https://github.com/varunvssrivastava/program6.1.git (fetch)
origin https://github.com/varunvssrivastava/program6.1.git (push)
• lrv24mc113_varun@varun-srivastava:~/DevOpsLab/program-6.1$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Dockerfile
        app.py
        package.json
        requirements.txt

nothing added to commit but untracked files present (use "git add" to track)
• lrv24mc113_varun@varun-srivastava:~/DevOpsLab/program-6.1$ git add .
• lrv24mc113_varun@varun-srivastava:~/DevOpsLab/program-6.1$ git commit -m "Initial commit"
[main (root-commit) f553e14] Initial commit
 4 files changed, 28 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 app.py
 create mode 100644 package.json
 create mode 100644 requirements.txt
• lrv24mc113_varun@varun-srivastava:~/DevOpsLab/program-6.1$ git push -u origin main
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (6/6), 780 bytes | 780.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/varunvssrivastava/program6.1.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
• lrv24mc113_varun@varun-srivastava:~/DevOpsLab/program-6.1$
```

Output:

varunsrivastava15/program6.2

Last pushed 13 minutes ago · ☆0 · ↓4

Add a description

Add a category

General Tags Image Management BETA Collaborators Webhooks Settings

Tags

This repository contains 1 tag(s).

Tag	OS	Type	Pulled	Pushed
latest		Image	less than 1 day	14 minutes

[See all](#)

Repository overview INCOMPLETE

DOCKER SCOUT INACTIVE
[Activate](#)



Build with
Docker

Accelerat

Docker Bu
dedicated

Get faster
support, s

Costs Res

 main  1 Branch  0 Tags

Add file

<> Code

 varunvsSrivastava Initial push 3924898 · 13 minutes ago  1 Commit		
 Dockerfile	Initial push	13 minutes ago
 app.py	Initial push	13 minutes ago
 requirements.txt	Initial push	13 minutes ago

 **README**

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